

Introduction

Two-frequency heating of ECR ion sources has been shown to produce 2-5 times more high-charge state ions versus single frequency heating, while also enhancing plasma stability and shifting the charge state distribution to higher charge state ions [2].

In this experiment, we explore the effects of 18 and 28 GHz heating on the charge state distributions.

Methodology

Axial x-rays emitted from the plasma travel through the 10 mm diameter extraction aperture and two collimators to be received by the XR-100T CdTe detector (Figure 1). A custom alignment arm (Figure 2) was created to hold the two collimators in place relative to each other while allowing for fine adjustments in all three directions. Their alignment was tested with a laser held approximately 2.5 m upstream of the Cu collimator to simulate x-rays coming from VENUS' extraction. The spacing between the collimators was chosen such that the field of view of the x-ray detector matches the 10 mm diameter of the extraction aperture (how big??), so that we only see x-rays emitted directly from the plasma.

For every run, VENUS was tuned primarily with the bias disk and injection pressure to optimize stability and production of O6+ on the faraday cup after the analyzing magnet. The magnetic fields were held constant (Figure 3).

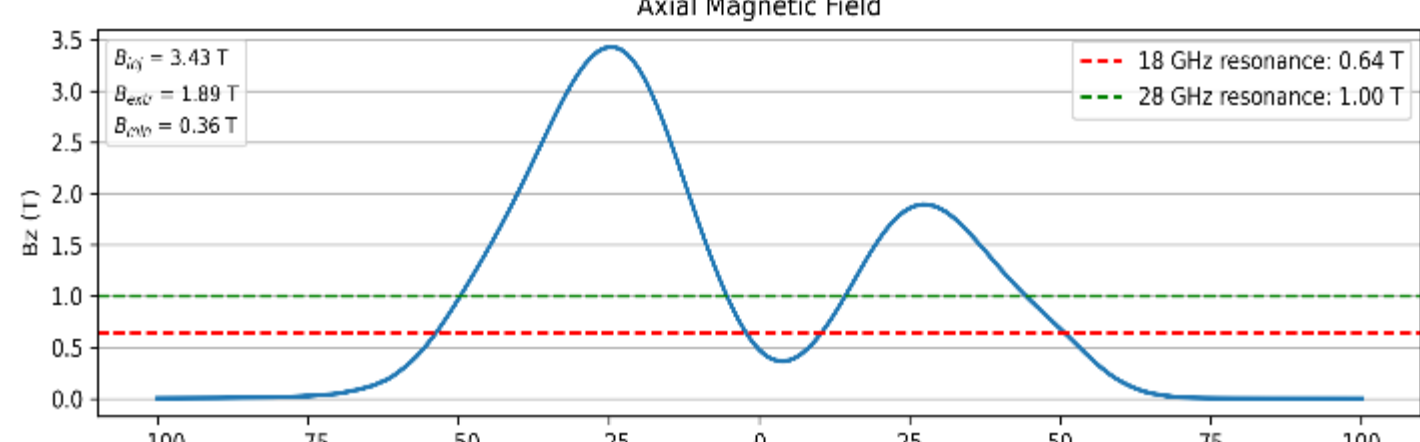


Figure 3: Axial magnetic field.

X-ray data were collected for 10 minutes at each power level (a combination of either 18 GHz klystron, 28 GHz gyrotron, or both) at various total power levels, and then a charge-state distribution (CSD) was taken (Figure 4).

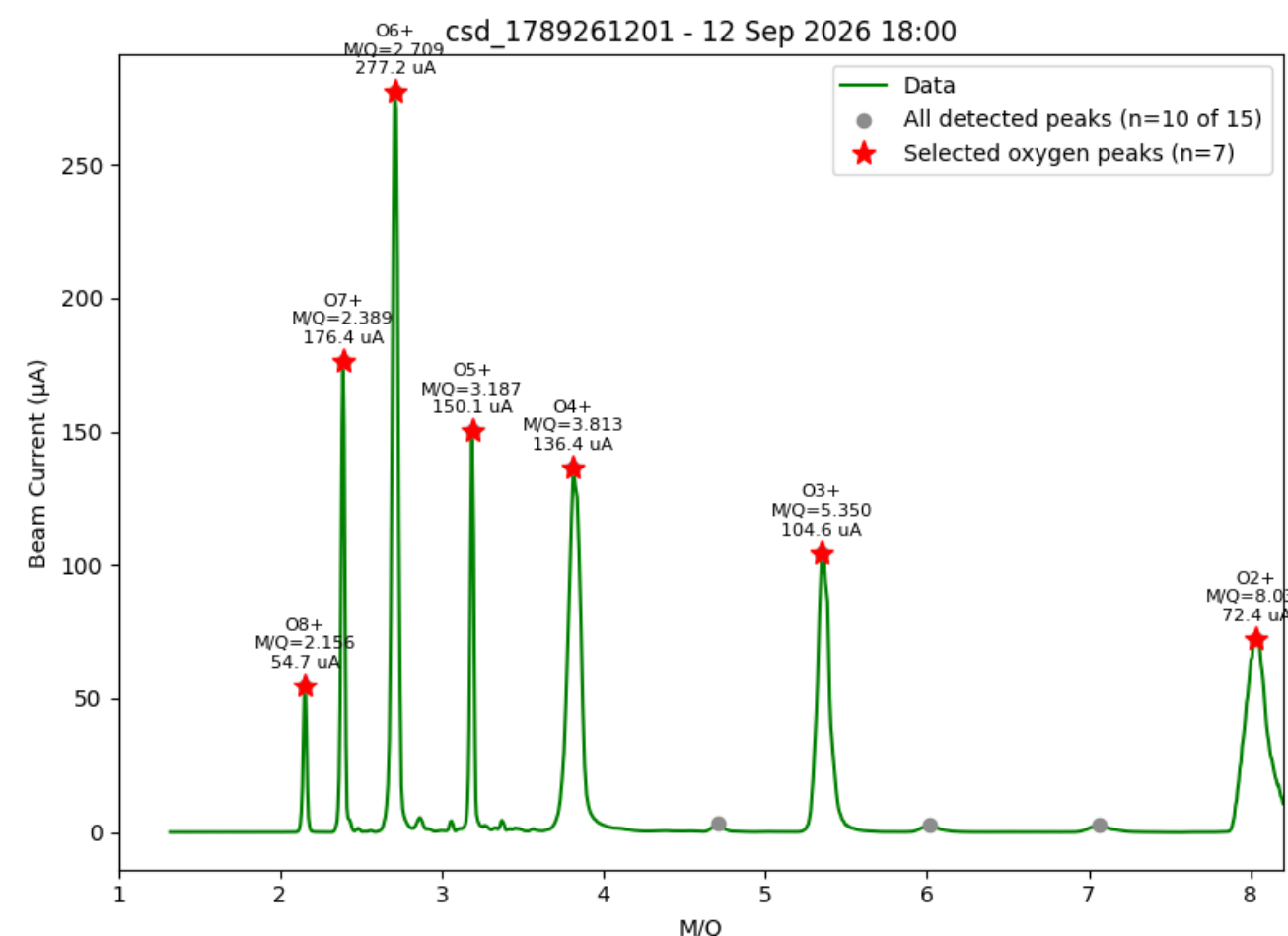


Figure 4: Sample CSD.

X-ray data was analyzed to find the spectral temperature (Figure 5) by taking the negative slope of the log of the normalized and detector-efficiency-corrected data (in the linear region). The steeper this slope is, the higher the hotter the electrons in the plasma are, which leads to a greater heat load on the superconducting magnet cryostat.

Formula for spectral temperature

Formula for beam current as function of frequency^2

Experimental Setup

Figure 1: Schematic of the XR-100-CdTe detector at the VENUS ion source.

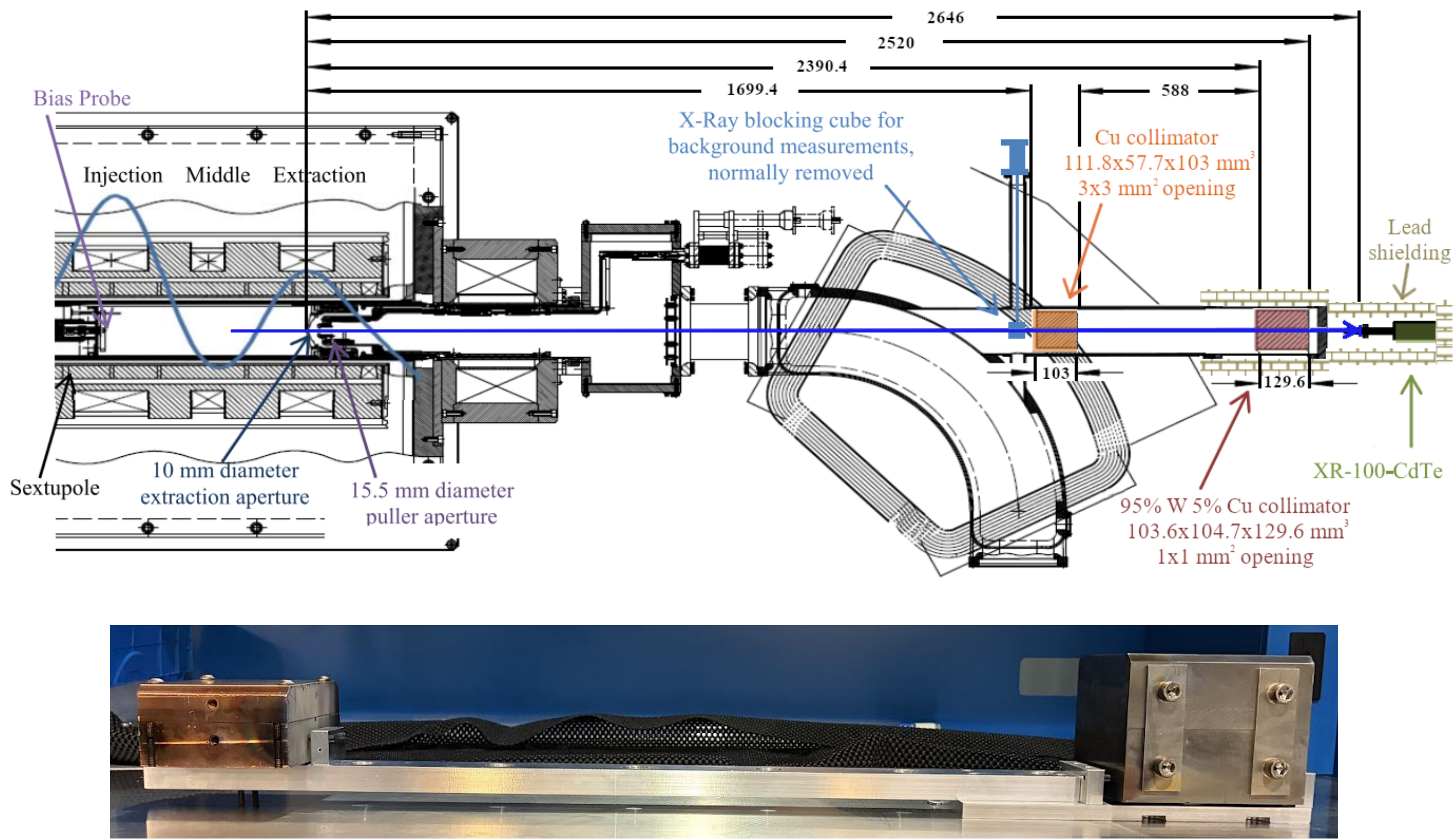


Figure 2: An adjustable collimator alignment arm, used to align and fix both collimators in place with the center of the beamline.

Two-Frequency Power Results

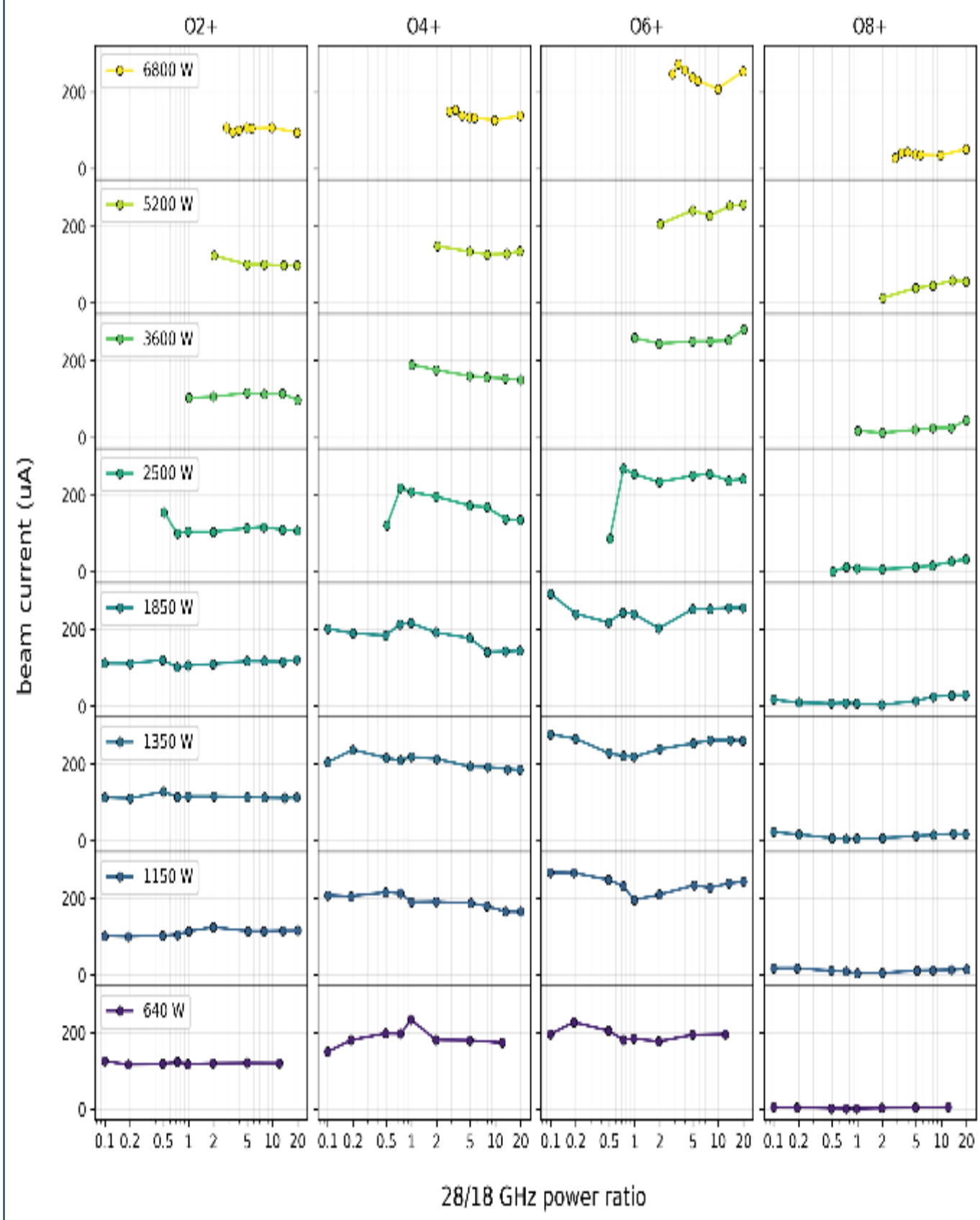
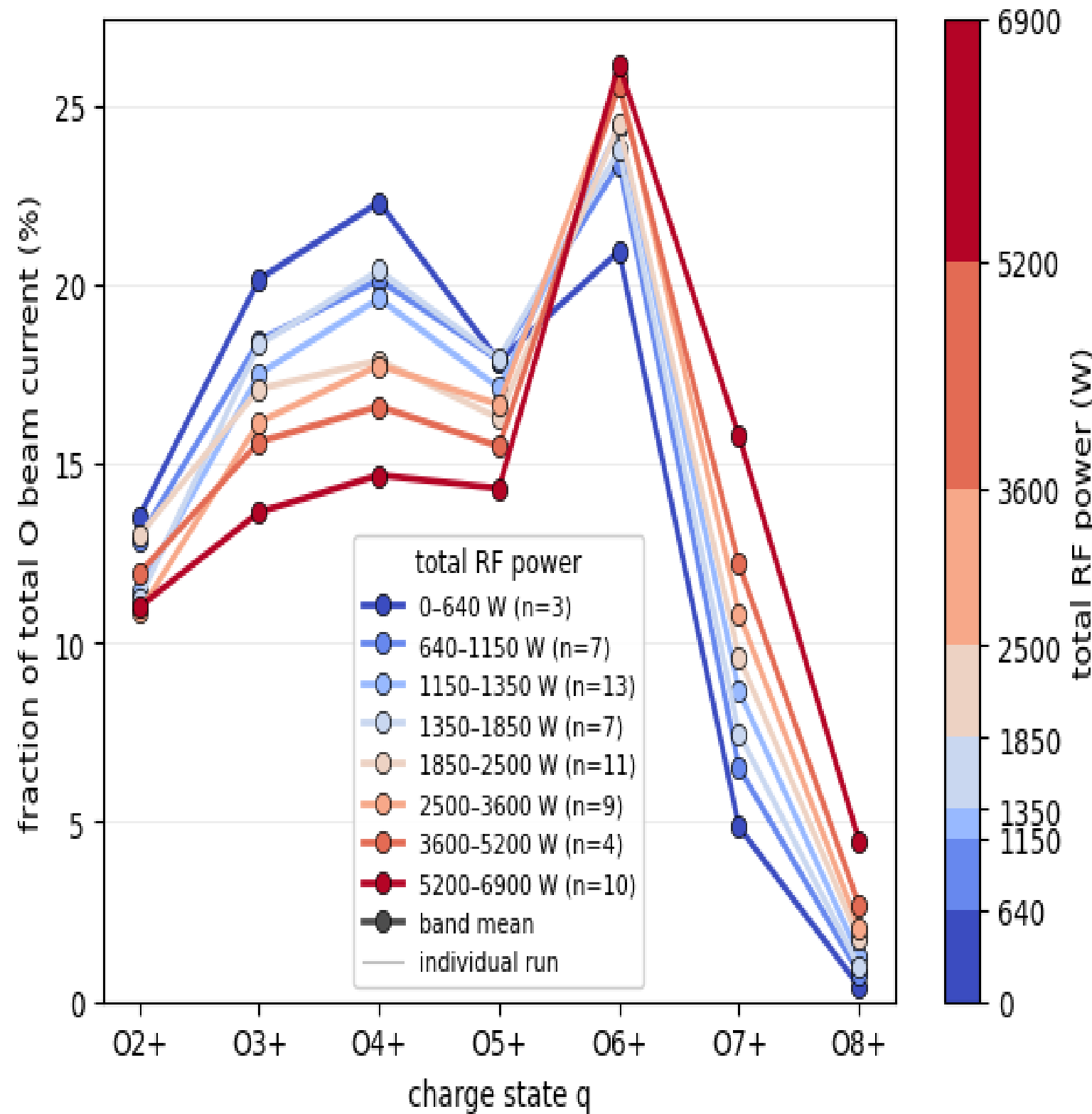


Figure 7: The oxygen charge state distribution shifts to higher charge state with more total power.

Figure 6: Beam currents of oxygen charge states vs 28 GHz to 18 GHz power ratio.



Comparison to Previous Work

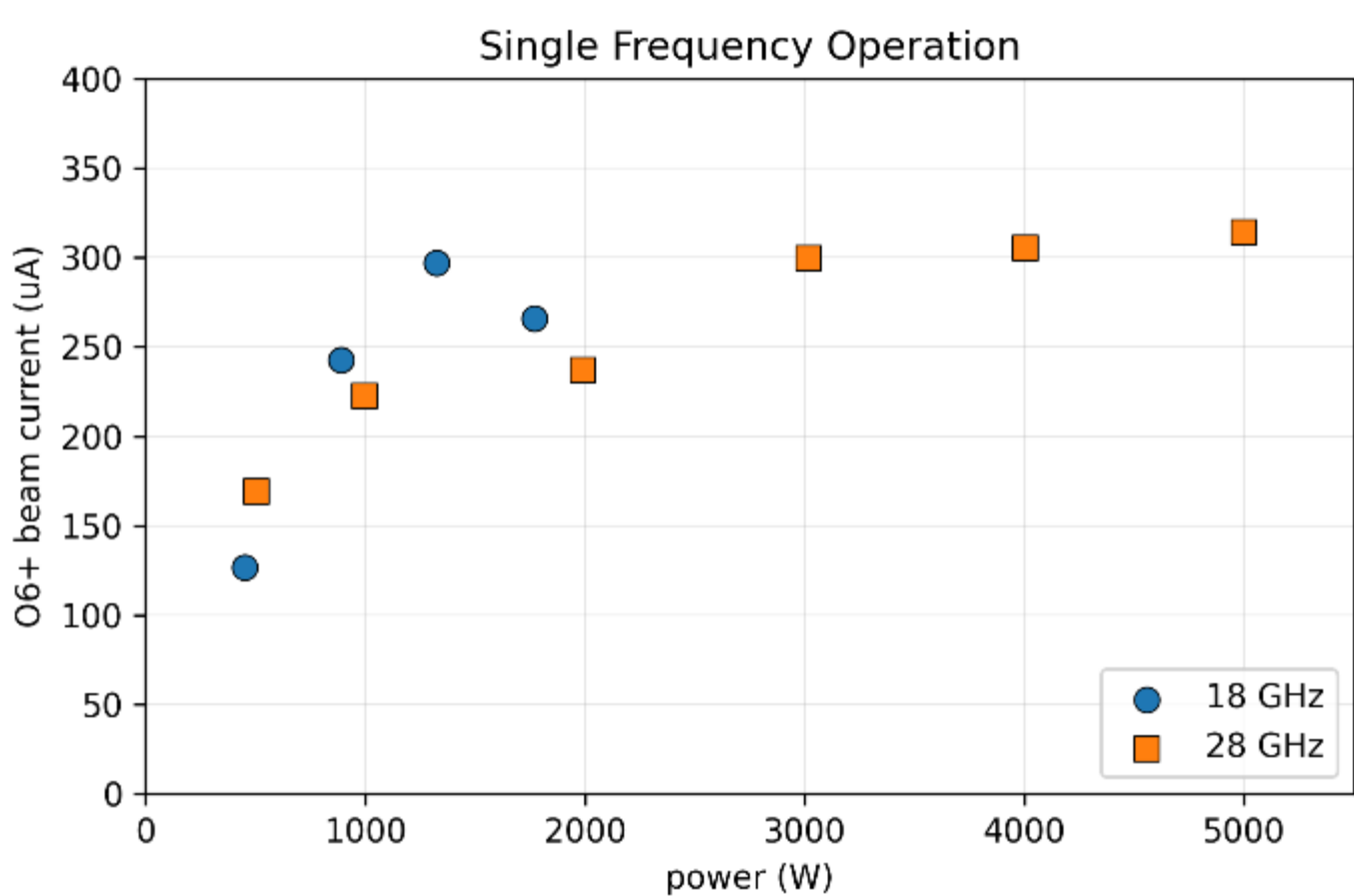


Figure 8: Single frequency heating agrees with previous measurements [4]. Beam current increases with total power.

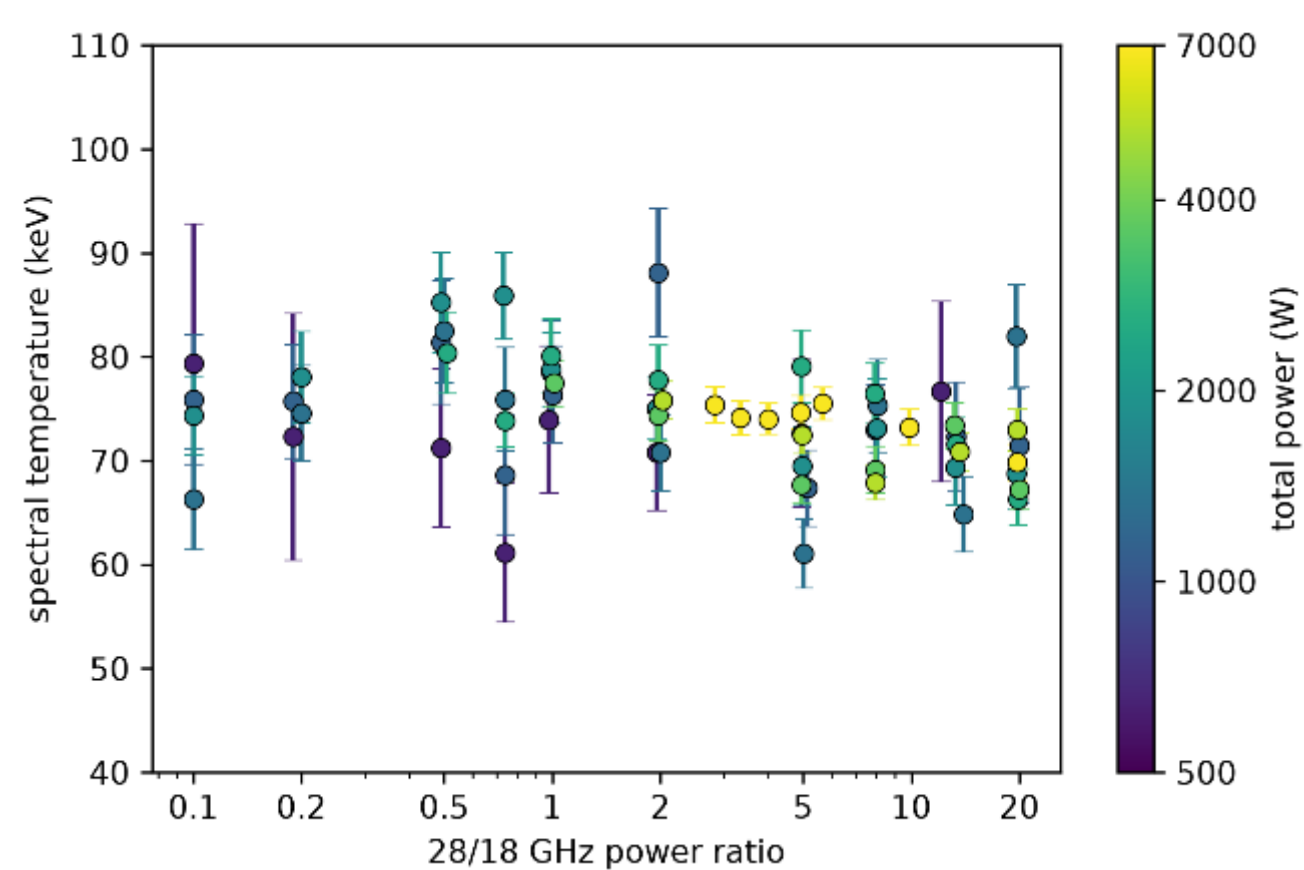


Figure 9: Spectral temperature agrees with previous measurements [5]. Does not change with different power or frequencies.

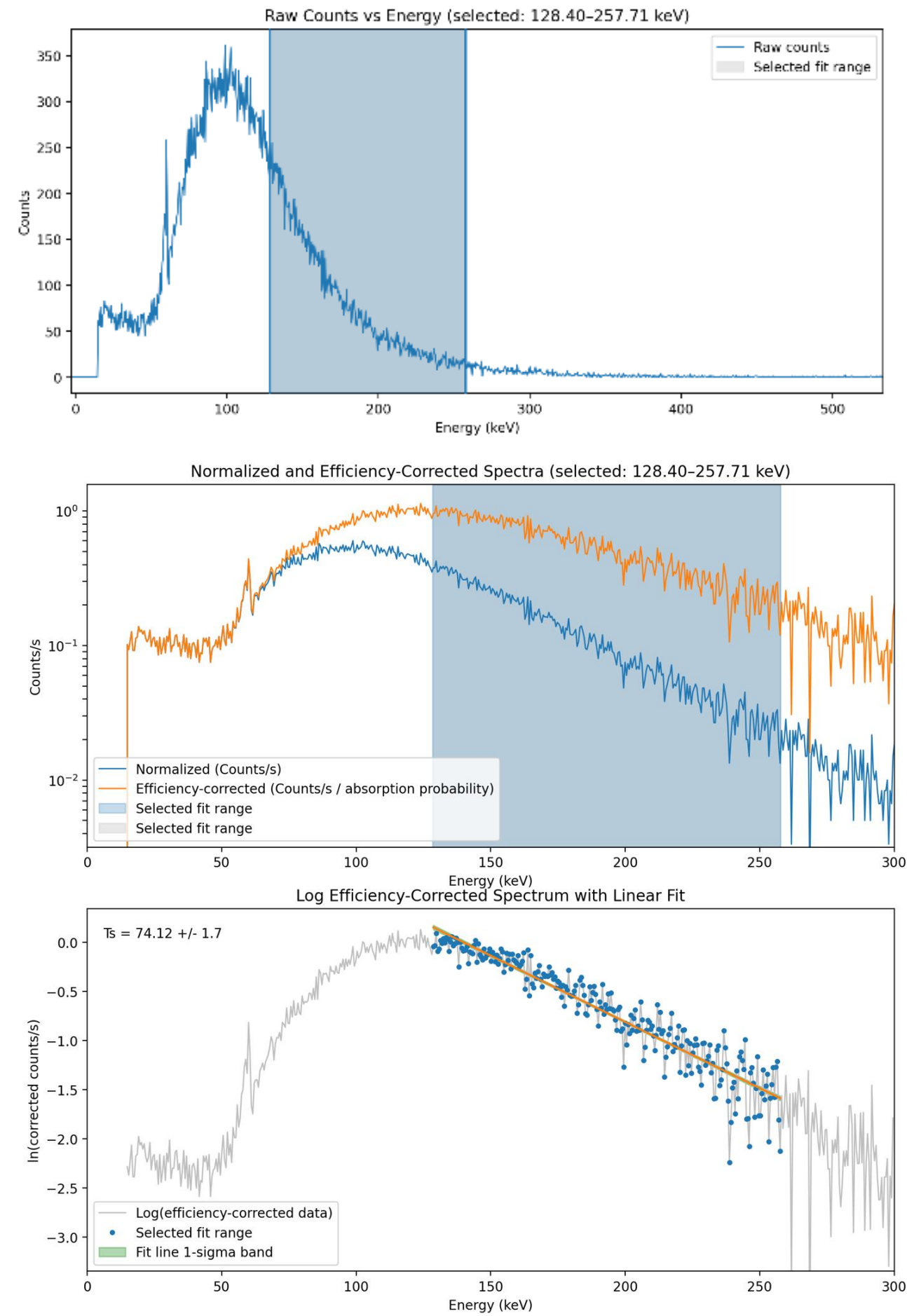


Figure 5: finding spectral temperature from x-ray spectrum.

Conclusion

At a constant magnetic field configuration, the oxygen beam currents do not depend on the ratio of 28 to 18 GHz power. Rather, the charge state distribution shifts to favor higher charge states with higher total power.

We would expect the beam current to increase as the square of the frequency, which does occur with single frequency heating (Figure 8). With two frequencies, however, this effect is negated.

Future Work

- Try different magnetic field configurations
- Developing a machine learning framework to predict behavior at higher total powers/higher frequencies
- Three frequency mixing

References

[1] Lyneis, Xie, 1995 12th International Workshop on ECR Ion Sources
[2] R Geller (should be first)
[4] Kitagawa, Raczk, Kato, Takasugi
[5] Benitez (only depends on Bmin)
[9] J Benitez, C Lyneis, L Phair, D Todd and D Xie 2017 IEEE Transactions on Plasma Science 45 7